YASHWANTH GOWDA

yashgangatkar@gmail.com | 91-9449273299 | LinkedIn | Portfolio | GitHub | Tempe, AZ, USA

PROFESSIONAL SUMMARY

Robotics and autonomous systems engineer with an MS from Arizona State University (May 2026, GPA: 3.47/4.0) and a BEng in Mechanical Engineering. Experienced in ROS2 architecture design, autonomous navigation, multi-sensor fusion, dexterous manipulation, and machine learning for robotics across both simulation and live hardware deployment.

Contributed to real hardware research at ASU's robotics lab, deploying control architectures on a 7-DOF tendon-driven robotic hand, achieving a 25% reduction in trajectory error recurrence across 20 test runs. Additional background in mechanical design (CATIA V5, SolidWorks, ANSYS) and cybersecurity governance, risk, and compliance (ISO 27001, NIST CSF, GDPR).

STEM OPT eligible. Seeking full-time roles in robotics engineering, autonomous systems, automation, systems engineering, and GRC — US-based or India-based.

EDUCATION

Master of Science in Robotics and Autonomous Systems (MAE) — *August 2024 – May 2026*
Arizona State University, Tempe, AZ, USA — GPA: 3.47/4.0 (4.0 — Spring 2026)

Coursework: Modeling and Control of Robots, Multi-Robot Systems, Perception in Robotics, Reinforcement Learning for Robotics, Applied Machine Learning for Mechanical Engineers, Linear Algebra in Engineering, Experimental & Deployment of Robotic Systems, Advanced System Modeling & Dynamics, Information Assurance and Security, and Technology Innovation Lab

Bachelor of Engineering in Mechanical Engineering — *September 2018 – July 2023*
Bangalore Institute of Technology (VTU), Bengaluru, India — CGPA: 7.13/10 (First Class)

Coursework: Finite Element Methods, Design of Machine Elements, Computer Aided Design and Manufacturing (CAD/CAM), Heat Transfer, Fluid Mechanics, Mechanics of Materials, Additive Manufacturing, Thermodynamics, Turbo Machines, Kinematics of Machines, Control Engineering, Metal Casting and Welding, Non-Destructive Testing

TECHNICAL SKILLS

Robotics and Autonomy:
ROS2, ROS, SLAM, Autonomous Navigation, Multi-Sensor Fusion (LiDAR, IMU, Depth Camera), Nav2, PID Control, Joint-Space Control, Inverse Dynamics, Robotic Manipulation, Swarm Robotics, OpenCV, Gazebo, RViz, Stretch3 Robot, Control Systems, Embedded Systems, Arduino

Programming and Machine Learning:
Python, C/C++, PyTorch, MATLAB, Simulink, Linux/Bash, Git/GitHub, Java (OOP), JavaScript

Mechanical Design and Simulation:
SolidWorks, CATIA V5, Solid Edge, ANSYS (FEA/CFD), CAD/CAM, CNC, Additive Manufacturing

Cybersecurity and GRC:
ISO 27001, NIST CSF, EU GDPR, MITRE ATT&CK, Risk Assessment, Incident Management, Security Auditing, Patch Management, Phishing Attack Monitoring, Microsoft Defender 365, KDMARC, SAFEScore.ai, Infosec IQ

Tools and Platforms:
LaTeX, Microsoft Office (Word, Excel, PowerPoint, Outlook),

Professional Skills:
Technical Documentation, Cross-functional Collaboration, Project Management, Client Communication, Adaptability, Problem Solving, Time Management

WORK EXPERIENCE

Research Assistant — *February 2026 – Present*
Robotics Lab — Prof. Zhe Xu, Arizona State University — Tempe, AZ

- Developed and deployed Python-based control architectures via ROS2 topics and services for the TetherIA Aero Hand, a 7-DOF tendon-driven robotic hand targeting dexterous manipulation tasks.
- Implemented and validated custom joint-space trajectory planners across 4 grasp configurations — rolling dice, bottle grasping, card picking, and teleoperation-based manipulation — achieving a 25% reduction in trajectory error recurrence across 20 test runs through iterative refinement.

- Integrated the TetherIA Aero Hand with a Stretch3 mobile manipulator via USB, enabling combined mobile and dexterous manipulation tasks — whole-body reach with finger-level grasping in a single coordinated system.
- Stack: Python, ROS2 (topics/services), Serial API, 7-DOF tendon-driven kinematics, joint-space control, Stretch3 mobile manipulator, USB hardware integration.

Governance Risk and Compliance Intern *March 2022 – January 2023*
CyRAAC Services Pvt. Ltd., Bengaluru, India
- Supported ISO 27001-aligned security governance activities across client environments, contributing to improved control alignment, audit readiness, and risk visibility.
- Assisted end-to-end information security audits, delivering cross-functional risk mitigation strategies that strengthened system resilience for critical infrastructure clients.
- Deployed and monitored Microsoft Defender 365, KDMARC, SAFEScore.ai, and Infosec IQ to proactively detect phishing attempts and manage patch vulnerabilities across regulated environments.

Office Assistant (Part-Time) *January 2019 – July 2024*
Sapthagiri Group, Bengaluru, India
- Managed end-to-end document workflows, meeting coordination, and inter-team communication across daily operations, performed concurrently with full-time academic study.
- Improved operational efficiency through systematic data entry, inventory tracking, and vendor coordination, contributing to smoother procurement cycles.

ACADEMIC PROJECTS

PhishLens — Neural Phishing and Malicious URL Detection System (GitHub) *January 2026 – May 2026*
Arizona State University — CSE 543: Information Assurance and Security
- Designed and trained a PyTorch character-level 1D CNN for malicious URL classification, mapping raw URLs into embedding representations and using convolutional filters with adaptive max-pooling to capture URL-level spatial patterns.
- Achieved 92.98% binary accuracy (ROC-AUC 0.9636), Multiclass Macro F1 of 0.9145, ROC-AUC of only −0.0288 drop under full adversarial attack, and 0.9929 ROC-AUC on 3 unseen datasets.
- Stack: PyTorch, 1D CNN, Character-level embeddings, Python REST API, JavaScript, Deep Learning, Adversarial ML, Cybersecurity, URL classification.

PAPI Controller: Solar Irrigation System (GitHub) *August 2025 – December 2025*
Arizona State University — Advanced System Modeling and Dynamics
- Designed a Two-Degree-of-Freedom (2-DOF) hybrid controller for solar-powered irrigation pump systems, combining a physics-based inverse dynamics feedforward layer with a feedback PID loop to decouple setpoint tracking from disturbance rejection.
- Achieved 60% faster rise time and near-instant disturbance rejection compared to a standard PID baseline, validated across 25 MATLAB/Simulink simulation test cases under variable solar irradiance and load disturbance conditions.
- Stack: MATLAB, Simulink, 2-DOF control architecture, inverse dynamics, PID, control systems theory.

ProphetLSTM-GAN: Rocket Engine Health Monitoring (GitHub) *August 2025 – December 2025*
Arizona State University — Applied Machine Learning for Mechanical Engineers
- Developed an unsupervised anomaly detection framework for liquid rocket engine telemetry using a hybrid LSTM-GAN architecture, trained on publicly available rocket engine time-series datasets (Kaggle) without requiring labelled fault data.
- Engineered a combined fault scoring metric fusing LSTM reconstruction error with GAN discriminator confidence, improving fault detection AUC by 3.7% over a standalone LSTM baseline — enabling earlier identification of developing engine faults.
- Stack: PyTorch, LSTM, GAN, Python, time-series analysis, anomaly detection, unsupervised learning.

Autonomous Warehouse Patrolling Robot (GitHub) *January 2025 – May 2025*
Arizona State University — Experimental and Deployment of Robotic Systems
- Designed and deployed a scalable, low-cost autonomous patrol robot for structured warehouse environments using a layered ROS2 control architecture with Nav2-based autonomous navigation and real-time SLAM-based mapping and localization.
- Engineered a multi-sensor fusion pipeline integrating LiDAR, IMU, and Depth camera data streams, enabling robust obstacle detection and continuous autonomous patrol without human intervention.
- Stack: ROS2, Nav2, SLAM, LiDAR, IMU, Depth Camera, Python, Gazebo, RViz, multi-sensor Fusion.

Debris Detection using Swarm Robots (GitHub) *August 2024 – December 2024*
Arizona State University — Multi-Robot Systems

- Designed and simulated a decentralized swarm robotics system of 10–15 autonomous agents for cooperative debris detection and mapping across a simulated environment in MATLAB.
- Implemented SLAM algorithms across the multi-agent system, enabling distributed navigation, obstacle avoidance, and collaborative maps merging without any central controller, achieving full area coverage through autonomous task partitioning and agent consensus.
- Stack: MATLAB, SLAM, swarm robotics, decentralized control, multi-agent systems.

Microcontroller-Based Line Follower AGV (GitHub) *August 2021 – August 2022*
Bangalore Institute of Technology - Physical model built and tested
- Designed and physically built a small-scale automated guided vehicle (AGV) for factory floor material transport, integrating IR sensor arrays with a PID-based motor control algorithm for smooth, real-time path following at variable speeds.
- Successfully tested the physical model under variable load and speed conditions, validating sensor response latency and PID tuning for reliable line tracking.
- Stack: Arduino, C, PID control, IR sensors, embedded systems, mechanical design.

Effect of Inclination on Natural Convection in Porous Enclosures (GitHub) *February 2021 – August 2021*
Bangalore Institute of Technology — 3rd Year Project
- Performed computational thermal analysis using ANSYS to investigate the effect of enclosure inclination angle on natural convection heat transfer characteristics in porous media.
- Simulated multiple inclination configurations, analyzing Nusselt number variation, velocity profiles, and temperature distribution patterns across the porous domain.
- Stack: ANSYS (FEA/CFD), thermal analysis, heat transfer, fluid mechanics, computational simulation.

CERTIFICATIONS

CATIA V5 — Certificate of Competency *September 2021*
Govt. Tool Room and Training Centre (GT&TC), Bengaluru, India (Reg. No.: ST160118399)
- 1-month hands-on competency program in 3D modelling, part design, assembly design, and technical drafting.

Java with OOP — Certificate of Internship *August 2020*
IC Solutions, Bengaluru
- 1-month internship on object-oriented programming and practical application development using Java.

COMMUNITY ENGAGEMENT AND EXTRACURRICULAR

Social Impact Volunteer — UN SDG Programmes *May 2021 – May 2022*
Cognition — Social Innovation and Research Centre, Bengaluru, India
- Completed **240 certified volunteer hours** across five UN SDG-aligned community programmes, supporting quality education, zero hunger, public health awareness, cleanliness initiatives, and financial inclusion through digitized transactions.

Volunteer — International Students and Scholars Center *June 2025 – May 2026*
Arizona State University
- Assist with organizing and running campus events for the international student community at Arizona State University.

Volunteer — Southwest Robotics Symposium (SWRS) 2024 *October – November 2024*
Ira A. Fulton Schools of Engineering, ASU
- Supported event logistics for a regional robotics research symposium, assisting researchers, attendees, and event organizers.

Event Coordinator / Backstage Volunteer *August 2018 – December 2021*
Bangalore Institute of Technology
- Coordinated backstage operations for Kannada Kalarava cultural festivals in 2018, 2020, and 2021.
- Supported event coordination for Manthan, the college festival at Bangalore Institute of Technology, in 2019.

LANGUAGES

English — Professional, Kannada — Native, Hindi — Fluent